Evaluation of (probably) obsolete modeling functions

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2026-09-02

Contents

1 Goal

The goal is to make the fgeo2 body of code lighter and easier to maintain by removing obsolete modeling functions. Statistical modeling requires flexibility and is better handled with maintained, dedicated tools. The new distribution of code (fgeo2) focuses on metrics, statistics, data handling, etc., leaving modeling aside for the most part.

This evaluation focuses on:

- primarily, removal of modeling functions;
- minimum edits to the remaining ones; and
- identifying dependents somewhere else, so all code remains working.

In this document I will often refer to the old script-based distribution of code as "CTFS", and the newer formal R packages as `ctfs` (a more formal distribution of CTFS code) and `fgeo` (same goals, different structure, heavy reorganization).

2 Candidate functions

In CTFS code, there are several blocks, with many functions each, that seem customized implementations of Bayesian models and/or legacy code for particular projects. It does not seem suitable for general use, and many of the functions are clear candidates for removal.

An earlier function-by-function evaluation was prepared during the transition from the CTFS scripts to the `ctfs` and `fgeo` packages. That discussion involved Sean McMahon, Stuart Davies, Mauro Lepore and myself (Gabriel Arellano). The results of that discussion were stored in a table edited directly by McMahon and Arellano; see `evaluation/old-evaluation-table.xlsx`. I have used also the the ctfs development notes, growth tests, and legacy regression tests. I have complemented that old review with a new review of the code, some tests, evaluation of dependencies, scientific need, and maintained alternatives. The table below is the consolidated evaluation of functions involved in Bayesian, pseudo-Bayesian, or specialized model fitting. Other functions that are also clear candidates for removal will be reviewed somewhere else.

Mod.	Script	Function	Notes	Recommendation
01	utilities-lmerBayes.r	lmerBayes	Fits a nonlinear hierarchical model with one grouping factor by running a hand-written Metropolis-within-Gibbs sampler. It has no explicit prior interface, keeps adapting proposal widths throughout the saved chain, provides no convergence diagnostics, and has only a disabled test that checked output names. Modern mixed-model and Bayesian packages cover this role.	remove
01	utilities-lmerBayes.r	lmerBayes.hyperllike.sigma	Calculates the multivariate-normal density of group parameters while lmerBayes updates one covariance element. It is only a technical helper for the hand-written covariance sampler.	remove
01	utilities-lmerBayes.r	lmerBayes.hyperllike.mean	Calculates the same group-parameter density while changing one hypermean. lmerBayes does not call it. This functionality is readily available from standard distribution functions.	remove
01	utilities-lmerBayes.r	full.likelihood.lmerBayes	Adds the group-level likelihoods used by lmerBayes and mixes observation likelihood with the density of random-effect parameters. The source itself questions what should be counted: it does not seem reliable.	remove
01	utilities-lmerBayes.r	llike.fixef.lmer	Replaces one fixed-effect value and recalculates the lmerBayes likelihood. It is only a parameter-update adapter for that sampler.	remove
01	utilities-lmerBayes.r	llike.model.lmer	Calculates observation likelihood for one group and adds the multivariate-normal density of that group's parameters. It contains browser calls for numerical failures and documents a Flat error option whose code is commented out. It is coupled to lmerBayes, not a reusable interface.	remove
01	utilities-lmerBayes.r	residual.llike.lmerBayes	Recalculates Gaussian or negative-binomial likelihoods across groups while one residual parameter is changed. It is only a technical helper for lmerBayes.	remove
01	utilities-lmerBayes.r	badSD	Returns whether a single residual parameter is non-positive. This one-line check is needed only by the old sampler.	remove
01	utilities-lmerBayes.r	arrangeParam.llike.2D	Copies a matrix and replaces one element, optionally mirroring it to keep a covariance matrix symmetric. It exists only to pass scalar proposals to the lmerBayes sampler.	remove
01	utilities-lmerBayes.r	arrangeParam.Gibbs.2D	Combines already-updated and not-yet-updated elements of two covariance matrices during a Gibbs sweep. It is bookkeeping specific to lmerBayes.	remove
01	utilities-lmerBayes.r	saveParamFile	Flattens each block of the lmerBayes parameter array and appends it to a tab-separated file to reduce memory use during long runs. This is an old storage workaround.	remove

01	utilities-lmerBayes.r	restoreParamFile	Reads the tab-separated chain written by saveParamFile, it has no use outside that storage workaround.	remove
01	utilities-lmerBayes.r	resetParam	Creates an empty parameter-array block after the preceding block has been written to disk, retaining only its last state. It is bookkeeping for saveParamFile.	remove
01	utilities-lmerBayes.r	summaryMCMC	Calculates posterior means, quantiles, predictions, and DIC from lmerBayes output, but not other standard diagnostics like effective sample size, R-hat, or multi-chain diagnostics. Its DIC depends on the not-fully-reliable combined likelihood above.	remove
01	utilities-lmerBayes.r	recalculate.lmerBayesllike	Attempts to recompute likelihoods along a saved lmerBayes chain. It cannot work as documented because keep is never replaced by fit$keep and it passes the residual argument under the wrong name.	remove
01	utilities-lmerBayes.r	covTocorr	Converts a covariance matrix to correlations with nested loops. Equivalent to the standard stats::cov2cor.	remove
01	utilities-modelBayes.r	modelBayes	Fits a user-supplied mean and residual model without random effects using scalar random-walk Metropolis updates. It has no explicit priors, adapts proposal widths throughout the chain, leaves its update argument unused, stops in browser on some failures, and failed after an earlier repair attempt during development of ctfs/fgeo.	remove
01	utilities-modelBayes.r	residual.llike.modelBayes	Calculates Gaussian or negative-binomial likelihood while one residual-model parameter is changed. It is only a technical helper for modelBayes.	remove
01	utilities-modelBayes.r	summaryModelMCMC	Calculates means, quantiles, and fitted values from modelBayes draws. It has no convergence diagnostics. The source explicitly warns that its prediction-array construction will fail for some dimensions.	remove
01	utilities-statistics.r	regression.Bayes	Fits an ordinary Gaussian linear regression with a conjugate Gibbs sampler. Priors are implicit, inputs are weakly checked. Better maintained regression tools already provide this type of model with tests and diagnostics.	remove
01	utilities-statistics.r	Gibbs.regslope	Draws regression coefficients conditional on the residual variance for regression.Bayes, only a technical helper for that function.	remove
01	utilities-statistics.r	Gibbs.regsigma	Draws the residual variance from an inverse-gamma distribution for regression.Bayes. It is only a technical helper and hard-codes the implied prior through its shape and scale formula.	remove
01	utilities-statistics.r	Gibbs.normalmean	Draws a normal mean conditional on a supplied standard deviation, with an implicit flat prior. No function in R/new calls it.	remove
01	utilities-statistics.r	Gibbs.normalvar	Draws a normal variance from an inverse-gamma distribution and returns the arbitrary value 0.01 when the data have zero variance. Its only current use is updating diagonal variances inside lmerBayes.	remove

01	utilities-statistics.r	model.xy	Fits an arbitrary one-predictor mean and residual model with hand-written Metropolis updates. It supplies no prior density, adapts proposals throughout the chain. The code seems to contain some errors or inconsistencies. Only used for the growth-bin model.	remove
01	utilities-statistics.r	arrangeParam.llike	Copies a parameter vector and replaces the element currently proposed by a sampler. It is simple bookkeeping. The NULL case can fail.	remove
01	utilities-statistics.r	arrangeParam.Gibbs	Builds the parameter vector for one step of a Gibbs sweep from the previous and current rows of a chain matrix. It is bookkeeping for the hand-written samplers.	remove
01	utilities-statistics.r	llike.GaussModel	Calculates a Gaussian log-likelihood for model.xy after calling a supplied prediction function. It enters browser rather than returning a controlled error when the result is non-finite, and it is used only by the old growth-model chain.	remove
01	utilities-statistics.r	llike.GaussModelSD	Calculates the same Gaussian likelihood while residual-model parameters are updated. Its default permits residual standard deviations to approach zero, which can make the likelihood arbitrarily large. It is used only by model.xy.	remove
01	utilities-statistics.r	BadParam	Always returns FALSE. It is a placeholder callback for model.xy, not a parameter check.	remove
01	utilities-statistics.r	graph.modeldiag	Opens several X11 devices for trace and likelihood plots from model.xy. There seems to be errors / inconsistencies in the code. Maintained MCMC plotting packages are more reliable and complete.	remove
01	utilities-statistics.r	metrop1step	Performs one scalar random-walk Metropolis proposal and changes the proposal width after every acceptance or rejection. It seems non-standard MCMC machinery.	remove
01	utilities-statistics.r	metrop1step.discrete	Performs one Metropolis proposal among character states. Nothing in R/new calls it, it has no tests, and it adds no features beyond a short standard calculation.	remove
01	utilities-statistics.r	testmcmcfunc	Prints and returns a normal log-density for manual testing of the Metropolis code. Its own description says it is no longer used.	remove
05	demogchange-abund.fit.CTFS.r	model.littleR.Gibbs	Fits a paper-specific hierarchical model of species mortality and population-change rates between two censuses, allowing several symmetric and asymmetric distributions across species. The code has inconsistent calls to data vs. demog, and it does not seem it has been tested over generalized datasets. If someone wanted to replicate, they would have to re-implement from scracth in Stan or another maintained framework.	remove

05	demogchange-abund.fit.CTFS.r	full.abundmodel.llike	Combines the observation and across-species densities for the paper-specific abundance model. It counts some terms more than once and is not reliable. Used only by model.littleR.Gibbs().	remove
05	demogchange-abund.fit.CTFS.r	prob.N1	Calculates the likelihood contribution for the second-census abundance of one species. Just a helper for full.abundmodel.llike() and model.littleR.Gibbs().	remove
05	demogchange-abund.fit.CTFS.r	spmean.mort.abundGibbs	Calculates the likelihood contribution for mortality and survival of one species. Just a helper for full.abundmodel.llike() and model.littleR.Gibbs().	remove
05	demogchange-abund.fit.CTFS.r	hyper.abundGibbs	Calculates the across-species density of population-change rates. Just a helper for full.abundmodel.llike() and model.littleR.Gibbs().	remove
05	demogchange-abund.fit.CTFS.r	hyper.mortGibbs	Calculates the across-species density of mortality rates. Just a helper for full.abundmodel.llike() and model.littleR.Gibbs().	remove
05	demogchange-abund.fit.CTFS.r	bad.asympower.param	Checks the allowed parameter range for the asymmetric-power distribution. Just a parameter-checking helper for full.abundmodel.llike() and model.littleR.Gibbs().	remove
05	demogchange-abund.fit.CTFS.r	bad.asymexp.param	Checks that the scale parameters for the asymmetric exponential or Gaussian distribution are positive. Just a parameter-checking helper for full.abundmodel.llike() and model.littleR.Gibbs().	remove
05	demogchange-abund.fit.CTFS.r	fitSeveralAbundModel	Runs model.littleR.Gibbs for adjacent census pairs, the first and last census, and two DBH thresholds. It adds batching and file-sized output but no new statistical method; one historical test took about 20 minutes and no active automated test remains.	remove
05	demogchange-abund.fit.CTFS.r	graph.abundmodel	Plots observed and fitted distributions of species population change, optionally plots mortality against change, and lists the largest increases and losses. All inputs and uncertainty curves come from model.littleR.Gibbs, so the function has no use once that model is removed; the plots should be straightforward to recreate from any validated fit.	remove
05	demogchange-abund.fit.CTFS.r	find.xaxis.hist	Creates the x grid used by graph.abundmodel, split at the fitted distribution center. It is a small plotting helper with no independent use.	remove
05	demogchange-abund.fit.CTFS.r	abundmodel.fit	Evaluates the fitted across-species density for graph.abundmodel. The code contains errors or inconsistencies. It is otherwise only a plotting helper.	remove

05	demogchange-demogChange.r	lmerMortLinear	Does not fit a model; it converts a linear predictor for log mortality into survival probability over each census interval. It was written as a response function for lmerBayes, has no detected users outside the candidate code, and the equation is simple to state directly in a future model.	remove
05	demogchange-demogChange.r	lmerMortFixedTime	Does not fit a model; it assigns a separate log-mortality parameter to each integer category and converts it to interval survival. It is another response function for the obsolete modeling code, and maintained formula-based tools can express the same model directly.	remove
05	growth-growthfit.bin.r	extract.growthdata	Builds a species-level table of size and growth between two censuses. It may be worth to keep and repair .	keep
05	growth-growthfit.bin.r	run.growthbin.manyspp	Fits one to four piecewise growth models for each sufficiently abundant species and saves the growing result after every species. It assumes log-transformed DBH, has dataset-specific defaults, and only batches the failed growth-bin model; its historical tutorial call remains disabled.	remove
05	growth-growthfit.bin.r	run.growthfit.bin	Fits successive one- to four-bin models and uses each result to initialize the next. It is only an organizer for growth.flexbin and adds no calculation useful after that fitting method is removed.	remove
05	growth-growthfit.bin.r	growth.flexbin	Fits piecewise linear growth with free breakpoints, first with optim and then optionally with model.xy. The Gibbs route failed in historical tutorial testing, while the optim-only route later tries to read MCMC fields that do not exist. Maintained segmented-regression packages cover the same task more reliably.	remove
05	growth-growthfit.bin.r	llike.linearbin.optim	Splits a combined vector into piecewise-growth and residual parameters and returns the Gaussian log-likelihood for optim. It is only an adapter for growth.flexbin.	remove
05	growth-growthfit.bin.r	defineBinBreaks	Chooses starting breakpoints from quantiles, equal-width bins, or minimum-width and minimum-sample rules. These are starting-value heuristics for growth.flexbin, not a general binning function, and no other code uses them.	remove
05	growth-growthfit.bin.r	defineSDpar	Returns fixed fallback starting values for a two- or three-parameter residual model. The values are arbitrary and the function is used only by growth.flexbin.	remove
05	growth-growthfit.bin.r	enoughSamplePerBin	Checks whether every proposed size bin contains a minimum number of observations. It is a small validator used only to constrain growth.flexbin breakpoints.	remove

05	growth-growthfit.bin.r	wideEnoughBins	Checks whether every proposed size bin exceeds a fraction of the observed size range. It is a small validator used only to constrain growth.flexbin breakpoints.	remove
05	growth-growthfit.bin.r	bad.binparam	Checks that growth-model breakpoints lie within the size range and meet the minimum width and sample rules. Its interpretation of the parameter vector is specific to linearmodel.bin.	remove
05	growth-growthfit.bin.r	bad.binsdpar	Checks the breakpoint and positive scale of the three-parameter residual model used by growth.flexbin. It has no use outside that parameterization.	remove
05	growth-growthfit.bin.r	calculateBinModel.BIC	Calculates BIC from residual sum of squares as if errors were homoscedastic Gaussian. growth.flexbin fits a size-dependent residual model, so this value does not match the likelihood actually optimized.	remove
05	growth-growthfit.bin.r	calculateBinModel.DIC	Calculates DIC from the mean saved log-likelihood and the likelihood at the median parameter values. DIC requires a clearly defined likelihood and normally the posterior mean; this calculation inherits the defects of model.xy and is not reliable for comparing the bin models.	remove
05	growth-growthfit.bin.r	calculateBinModel.AIC	Returns log-likelihood minus twice the parameter count. Standard AIC is minus twice the log-likelihood plus twice the parameter count, so both the sign and scale are wrong.	remove
05	growth-growthfit.bin.r	calculateBinModel.bestpred	Tries to average predictions across saved MCMC draws but swaps the draw and parameter dimensions. It loops over parameters as if they were draws and therefore returns the wrong prediction.	remove
05	growth-growthfit.bin.r	assembleBinOutput	Tries to flatten the selected growth-bin fits into one species table. compare.growthbinmodel returns slopes, but this function first asks for slope, so it fails before assembling the table; the remaining column indexing is also tied to the old result layout.	remove
05	growth-growthfit.graph.r	graph.growthmodel.spp	Plots observed growth, a fitted piecewise curve, residual bands, and optional MCMC curves for one species. It expects fields that growth.flexbin does not always create, including meanpred. Too linked to the obsolete fit routines to keep.	remove
05	growth-growthfit.graph.r	graph.growthmodel	Loops over species and sends each growth.flexbin result to graph.growthmodel.spp, optionally opening a PDF or X11 device. It is only a batch plotting wrapper.	remove
05	growth-growthfit.graph.r	overlay.growthbinmodel	Overlays the one- to four-bin growth.flexbin curves for each species. It is only a plotting wrapper for those obsolete model objects.	remove

05	growth-growthfit.graph.r	compare.growthbinmodel	Ranks bin models with the incorrect AIC calculation, and it indexes fit$keep on the outer list instead of the individual fit when calculating mean likelihood. The selected model and returned slopes are therefore not reliable.	remove
05	growth-growthfit.graph.r	graph.outliers.spp	Compares full and trimmed growth tables, returns records absent after trimming, and plots them for one species. The useful part is data checking. Treatment and review for outliers should be probably part of the QA/QC routines.	remove
05	growth-growthfit.graph.r	graph.outliers	Calls graph.outliers.spp for many species and manages graphics devices. It adds no checking logic of its own.	remove
05	growth-growthfit.graph.r	binGraphSampleSpecies	Makes a fixed four-panel figure by retrieving fit objects from the global environment with get. It is tied to growth.flexbin output and to object names rather than explicit inputs.	remove
05	growth-growthfit.graph.r	binGraphManySpecies.Panel	Makes four panels by calling binGraphManySpecies for named global objects. It is an untested figure-layout wrapper with no analytical role.	remove
05	growth-growthfit.graph.r	binGraphManySpecies	Overlays selected growth curves from many species, but it first calls the broken comparison and assembly functions and retrieves data by name from the global environment. It cannot provide a trustworthy summary of the obsolete fits.	remove

2.1 Functions to be kept or reviewed

I keep `extract.growthdata()`, moving it to the growth-growth.r script. The rest of the candidates are removed.

3 Due diligence for dependents and dependencies

This due diligence has two main goals:

1. identify any function outside the candidate set that would break if a candidate were removed; and
2. identify non-candidate functions that may no longer be necessary if the functions marked **remove** are removed.

The dependency structure below uses every R script in the `R/new` directory of all seven fgeo2 modules. The exact pre-work snapshot is commit `040b73e`.

```
path <- normalizePath(system2("git", c("rev-parse", "--show-toplevel"),
                              stdout = TRUE)[1])

library(igraph, quietly = TRUE)
```

```
Warning: package 'igraph' was built under R version 4.4.3


Attaching package: 'igraph'

The following objects are masked from 'package:stats':

    decompose, spectrum

The following object is masked from 'package:base':

    union
```

```r
source(file.path(path, "evaluation/0-code-to-source.r"))

pre_work_commit <- "040b73e0cd9a0d55d64c1639b22159d5c6cbd20e"

modules <- c(
  "01_utilities", "02_biomass", "03_spatial",
  "04_abundance_diversity", "05_demography_growth",
  "06_topography_habitat", "07_mapping_plotting"
)

all_files <- system2(
  "git",
  c("-C", shQuote(path), "ls-tree", "-r", "--name-only", pre_work_commit),
  stdout = TRUE
)

module_pattern <- paste(modules, collapse = "|")
scripts <- all_files[
  grepl(paste0("^(", module_pattern, ")/R/new/.*\\.[Rr]$"), all_files)
]

code_env <- new.env(parent = .GlobalEnv)

for(file in scripts)
{
  code <- system2(
    "git",
    c("-C", shQuote(path), "show", paste0(pre_work_commit, ":", file)),
    stdout = TRUE
  )

  eval(parse(text = paste(code, collapse = "\n")), envir = code_env)
}

objects <- mget(
  ls(code_env, all.names = TRUE),
  envir = code_env,
  inherits = FALSE
)

all_functions <- names(objects)[vapply(objects, is.function, logical(1))]
```

```r
edges <- get_all_dependencies(envir = code_env)
edges <- edges[, c("from_function", "to_function"), drop = FALSE]
colnames(edges) <- c("from", "to")

candidates <- intersect(candidate_functions, all_functions)
removal_candidates <- intersect(removal_functions, all_functions)
missing_candidates <- setdiff(candidate_functions, all_functions)

if(length(missing_candidates))
  stop("Candidate functions not found: ", paste(missing_candidates, collapse = ", "))
```

3.1 Dependents that could break

For each candidate, the following code follows dependency arrows through all levels and retains only dependents outside the candidate set. These are the functions that would require attention before removal.

```r
dependent_rows <- list()

for(candidate in candidates)
{
  direct_dependents <- unique(edges$to[edges$from == candidate])
  direct_dependents <- setdiff(direct_dependents, candidates)

  all_dependents <- downstream_dependents(candidate, edges)
  all_dependents <- setdiff(all_dependents, candidates)

  if(length(all_dependents) == 0)
    next

  dependent_rows[[length(dependent_rows) + 1L]] <- data.frame(
    Candidate = candidate,
    `Direct dependents outside candidates` = paste(direct_dependents, collapse = ", "),
    `Dependents outside candidates at any level` = paste(all_dependents, collapse = ", "),
    check.names = FALSE
  )
}

if(length(dependent_rows) == 0)
{
  cat("**Result:** No function outside the candidate set depends on a candidate in the pre-w
```

```
} else {
  dependent_table <- do.call(rbind, dependent_rows)
  dependent_table_latex <- knitr::kable(
    dependent_table,
    format = "latex",
    booktabs = TRUE,
    longtable = TRUE,
    row.names = FALSE
  )

  dependent_table_latex <- sub(
    "\\begin{longtable}{lll}",
    "\\begin{longtable}{@{}L{4cm}L{9cm}L{9cm}@{}}",
    dependent_table_latex,
    fixed = TRUE
  )

  cat(
    "\\begin{landscape}\n",
    "\\footnotesize\n",
    "\\setlength{\\tabcolsep}{4pt}\n",
    dependent_table_latex,
    "\n\\end{landscape}\n",
    sep = ""
  )
}
```

Result: No function outside the candidate set depends on a candidate in the pre-work snapshot.

3.2 Dependencies that may become unnecessary

This check follows dependencies upstream through all levels. A non-candidate dependency is marked as potentially unnecessary when it belongs to the dependency chain of functions marked **remove** and has no downstream use outside that potential removal set. Functions marked **keep** or **review** prevent a shared dependency from receiving that conclusion.

```
candidate_dependencies <- character()

for(candidate in removal_candidates)
  candidate_dependencies <- c(
```

```
    candidate_dependencies,
    upstream_dependencies(candidate, edges)
  )

candidate_dependencies <- unique(candidate_dependencies)
candidate_dependencies <- setdiff(candidate_dependencies, candidates)
potential_removal_set <- c(removal_candidates, candidate_dependencies)

dependency_rows <- list()

for(dependency in candidate_dependencies)
{
  all_users <- downstream_dependents(dependency, edges)
  removal_users <- intersect(all_users, removal_candidates)
  other_users <- setdiff(all_users, potential_removal_set)

  conclusion <- "Still required elsewhere"
  if(length(other_users) == 0)
    conclusion <- "Potentially unnecessary"

  dependency_rows[[length(dependency_rows) + 1L]] <- data.frame(
    Dependency = dependency,
    `Functions marked remove downstream` = paste(removal_users, collapse = ", "),
    `Other downstream functions` = if(length(other_users)) paste(other_users, collapse = ", "
    Conclusion = conclusion,
    check.names = FALSE
  )
}

if(length(dependency_rows) == 0)
{
  cat("**Result:** No non-candidate fgeo2 dependencies were detected.\n")
} else {
  dependency_table <- do.call(rbind, dependency_rows)
  dependency_table <- dependency_table[
    order(
      dependency_table$Conclusion != "Potentially unnecessary",
      dependency_table$Dependency
    ),
  ]

  dependency_table_latex <- knitr::kable(
```

```r
    dependency_table,
    format = "latex",
    booktabs = TRUE,
    longtable = TRUE,
    row.names = FALSE
  )

  dependency_table_latex <- sub(
    "\\begin{longtable}{llll}",
    "\\begin{longtable}{@{}L{3.4cm}L{8.1cm}L{8.1cm}L{2.5cm}@{}}",
    dependency_table_latex,
    fixed = TRUE
  )

  cat(
    "\\begin{landscape}\n",
    "\\footnotesize\n",
    "\\setlength{\\tabcolsep}{4pt}\n",
    dependency_table_latex,
    "\n\\end{landscape}\n",
    sep = ""
  )
}
```

Result: No non-candidate fgeo2 dependencies were detected.

4 Current evaluation and decisions

The consolidated evaluation recommends removing 67 of the 73 candidates, keeping and repairing `extract.growthdata()` later. I created some issues for further development inspired by the obsolete code.

General Bayesian, hierarchical, segmented, and model-comparison tasks are better handled by maintained tools such as Stan, `brms`, etc. fgeo2 should not preserve it. The legacy implementations remain available under the `old` directories and in repository history.

5 Package structure after removals

The graph is intentionally deferred until after the removals. Once the work is complete, replace the post-work commit and date below and enable the chunk. The graph will use every toplevel function in the seven `R/new` directories, including functions with no detected internal dependency edges.

```
post_work_commit <- "REPLACE_WITH_POST_WORK_COMMIT"
post_work_date <- as.Date("YYYY-MM-DD")

post_all_files <- system2(
  "git",
  c("-C", shQuote(path), "ls-tree", "-r", "--name-only", post_work_commit),
  stdout = TRUE
)

post_scripts <- post_all_files[
  grepl(paste0("^(", module_pattern, ")/R/new/.*\\.[Rr]$"), post_all_files)
]

post_code_env <- new.env(parent = .GlobalEnv)

for(file in post_scripts)
{
  code <- system2(
    "git",
    c("-C", shQuote(path), "show", paste0(post_work_commit, ":", file)),
    stdout = TRUE
  )

  eval(parse(text = paste(code, collapse = "\n")), envir = post_code_env)
}

post_objects <- mget(
  ls(post_code_env, all.names = TRUE),
  envir = post_code_env,
  inherits = FALSE
)

post_functions <- names(post_objects)[vapply(post_objects, is.function, logical(1))]

post_edges <- get_all_dependencies(envir = post_code_env)
post_edges <- post_edges[, c("from_function", "to_function"), drop = FALSE]
colnames(post_edges) <- c("from", "to")

post_vertices <- data.frame(name = post_functions)
g.after <- graph_from_data_frame(
```

```
  post_edges,
  directed = TRUE,
  vertices = post_vertices
)

remaining_candidates <- intersect(candidate_functions, post_functions)
post_downstream <- character()

for(candidate in remaining_candidates)
  post_downstream <- c(post_downstream, downstream_dependents(candidate, post_edges))

post_downstream <- unique(post_downstream)
post_downstream <- setdiff(post_downstream, remaining_candidates)

V(g.after)$color <- "gray80"
V(g.after)$color[V(g.after)$name %in% post_downstream] <- "yellow"
V(g.after)$color[V(g.after)$name %in% remaining_candidates] <- "orange"

post_work_pdf <- file.path(
  path,
  "evaluation/results",
  paste0(
    "fgeo2-dependencies-", substr(post_work_commit, 1, 7), "-",
    format(post_work_date, "%Y-%m-%d"), ".pdf"
  )
)

dir.create(dirname(post_work_pdf), recursive = TRUE, showWarnings = FALSE)

set.seed(1)

pdf(post_work_pdf, height = 33, width = 33)
par(mar = c(0, 0, 2, 0))

plot(
  g.after,
  layout = layout.fruchterman.reingold,
  vertex.label = V(g.after)$name,
  vertex.size = 6,
  vertex.color = V(g.after)$color,
  vertex.frame.color = "white",
  vertex.label.color = "black",
```

```r
  vertex.label.cex = 2,
  edge.color = "gray45",
  edge.width = 0.7,
  edge.arrow.size = 0.3
)

legend(
  "topleft",
  legend = c(
    "Candidate remaining",
    "Downstream dependent",
    "Other function"
  ),
  pt.bg = c("orange", "yellow", "gray80"),
  pch = 21,
  pt.cex = 2,
  bty = "n"
)

dev.off()

post_work_pdf_relative <- file.path("results", basename(post_work_pdf))
knitr::include_graphics(post_work_pdf_relative)
```